

Jack Bulson

Gameplay Programmer

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Personal Profile

Gameplay / UI Programmer with over 4 years experience in the games industry. I enjoy creating fun gameplay mechanics, working collaboratively across disciplines and optimising experiences for both players and developers. Proficient in C++ and Unreal Engine, with experience shipping multiple titles.

Employment

Red Rover Interactive

Apr 2025 ~ Oct 2026

Senior Gameplay Programmer

Oct 2025 ~ Oct 2026

Gameplay Programmer

Apr 2025 ~ Oct 2025

Worked on **Enginefall**, a multiplayer PVP survival crafting/extraction game.

- Owned and implemented multiple systems. As part of this, handled scoping out the features, giving feedback on design, and iterating on them from player feedback.
- Improved building system; refactored to support prefabs (collections of building pieces), made placing objects easier and refactored build zones to improve UX for design.
- Fixed character collision bugs by refactoring and simplifying how character collision was handled. Implemented a component that managed collision profiles that sorted them by priority.
- Implemented a tech tree system and UI. This supported player or group progression depending on the tech tree. It was able to support large numbers of nodes across multiple pages.
- Created a world marker system which handled data from objects within the game for use on an in-game compass and map. This involved tracking various dynamic and static data sources across the server and client, ensuring that any replication cost was kept to a minimum.
- Refactored how item descriptions worked by implementing a stat system for items, that pulled stats dynamically from the item. This stopped text becoming out of date, improved the workflow as text no longer had to manually be updated, and made it easier to support localisation.
- Supported the programming team with my UI knowledge, offering advice and solutions in addition to preparing and delivering a presentation to give them an overview of UI in Unreal.

Sumo Newcastle

May 2022 ~ Mar 2025

Gameplay Programmer

Jan 2024 ~ Mar 2025

Graduate Programmer

May 2022 ~ Jan 2024

Worked on **Critter Cafe**, a single-player critter collecting, cafe management cozy game.

- Implemented multiple gameplay mechanics using the **Gameplay Ability System**, these included: character's three tools, blocks/redirectors/switches that sat on a grid-based puzzle system.
- Created these mechanics from a specification, and worked collaboratively with designers & level designers to iterate on them and develop tooling to solve workflow issues.
- Implemented the cafe gameplay in addition to UI and a minigame UI using **CommonUI**.
- Responsible for reworking the customisation mode to improve the game feel and UX.
- Experience gained with fixing bugs and debugging across a complex game with data used across a variety of places, such as save data or global game data.

Worked on **DeathSprint 66**, a multiplayer, high-speed on-foot racing game.

- Prototyped gamemodes and mechanics as the sole programmer within a small feature team, iterating on these to find the fun and moving on once we were satisfied with the outcome.
- Refactored game state and logic that was hard-coded for a race to allow for alternate gamemodes
- Implemented a dynamic team and team-scoring system, hit-effect system, end of match screens.
- Gained experience with networking, systems programming, physics and materials.

Worked on a shelved co-op multiplayer project.

- Implemented down-but-not-out functionality, gaining experience with the **Gameplay Ability System**, character animations. Added projected-to-screen UI indicators to highlight downed team-mates.
- Suggested and implemented outlines for the interaction system, improving the UX.

Personal Projects

Rendering Project (C++ / Vulkan)

2025

- Ported the rendering portion of my game engine to gain more experience with C++ (without Unreal)
- Implemented basic SSAO and created my own format that I could serialize assets into. This is so that they were optimised for the data formats within my project and easier to load at run-time.

Game Engine (Rust / Vulkan)

2022 ~ 2023

- Consisted of a renderer, simple asset management, collision detection and other utilities
- Implemented lighting, shadows, post-processing, instanced rendering, compute shaders
- Created a renderlist that used virtual resources to abstract away render passes

Technical Skills & Qualifications

- 4 years industry experience using C++/Unreal Engine/Perforce
- MComp Computer Science (Games Engineering) Newcastle University, UK (1st)

Other Activities & Interests

- I went back to Newcastle University in 2024 representing Sumo Newcastle to discuss what it's like entering the games industry and the expectations as a graduate on a day-to-day basis.
- I enjoy playing piano, bass guitar & writing my own music.